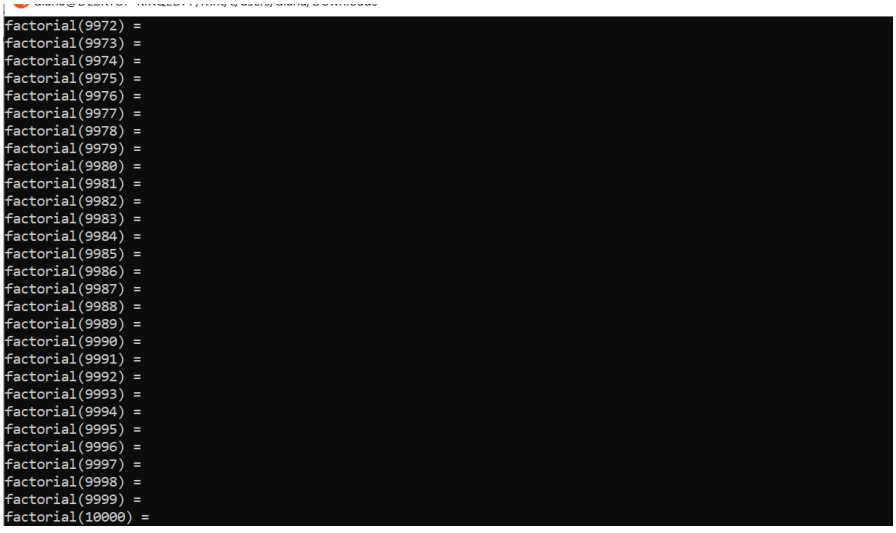
Coman Diana 343C5, lab 1

TASK 1

```
student@uso:~/Desktop/lab1$ vmstat -d | sort -nr -k 2
sda   178257  22558 9818402 4633524  95147 235258 34666256 15906656      0   1661
loop8  25241      0   54484  165996      0      0       0       0      0      5
loop7  10448      0   24976   63552      0      0       0       0      0      2
loop15   1181      0    3554    1552      0      0       0       0      0      0
loop6    989      0    6040    3648      0      0       0       0      0      0
loop1    841      0    2868    3072      0      0       0       0      0      0
loop16    797      0    5630    2128      0      0       0       0      0      0
loop14    763      0    5574    2360      0      0       0       0      0      0
loop2    546      0    5164    3248      0      0       0       0      0      0
loop9    507      0    2206     860      0      0       0       0      0      0
loop4    120      0    1432    1188      0      0       0       0      0      0
loop12    117      0     534    1648      0      0       0       0      0      0
loop0    111      0    1414    1404      0      0       0       0      0      0
loop5     93      0     498    1520      0      0       0       0      0      0
loop11     81      0     462     584      0      0       0       0      0      0
loop13     76      0    1328    1064      0      0       0       0      0      0
loop3     71      0     454    2076      0      0       0       0      0      0
loop10     38      0     184     736      0      0       0       0      0      0
sr0        0      0       0       0      0      0       0       0      0      0
disk- ------------reads------------ ------------writes----------- -----IO------
      total merged sectors      ms  total merged sectors      ms    cur    sec
```

TASK 2

```
student@uso:~/Desktop/lab1$ taskset -c $((seq 1 2 $(nproc)) | tr '\n', ','| sed 's/.$//')  ./fact_rcrs.py
10000
```



TASK 3

```
student@uso:~/Desktop/lab1$ for ((i = 0; i<10;i++)); do time zip -${i} -r ray-${i}.zip RAY.BMP
1>/dev/null; done

real    0m0,176s
user    0m0,000s
sys     0m0,007s

real    0m0,020s
user    0m0,015s
sys     0m0,004s

real    0m0,048s
user    0m0,024s
sys     0m0,000s

real    0m0,071s
user    0m0,033s
sys     0m0,000s

real    0m0,056s
user    0m0,028s
sys     0m0,000s

real    0m0,108s
user    0m0,043s
sys     0m0,000s

real    0m0,168s
user    0m0,070s
sys     0m0,000s

real    0m0,211s
user    0m0,103s
sys     0m0,000s

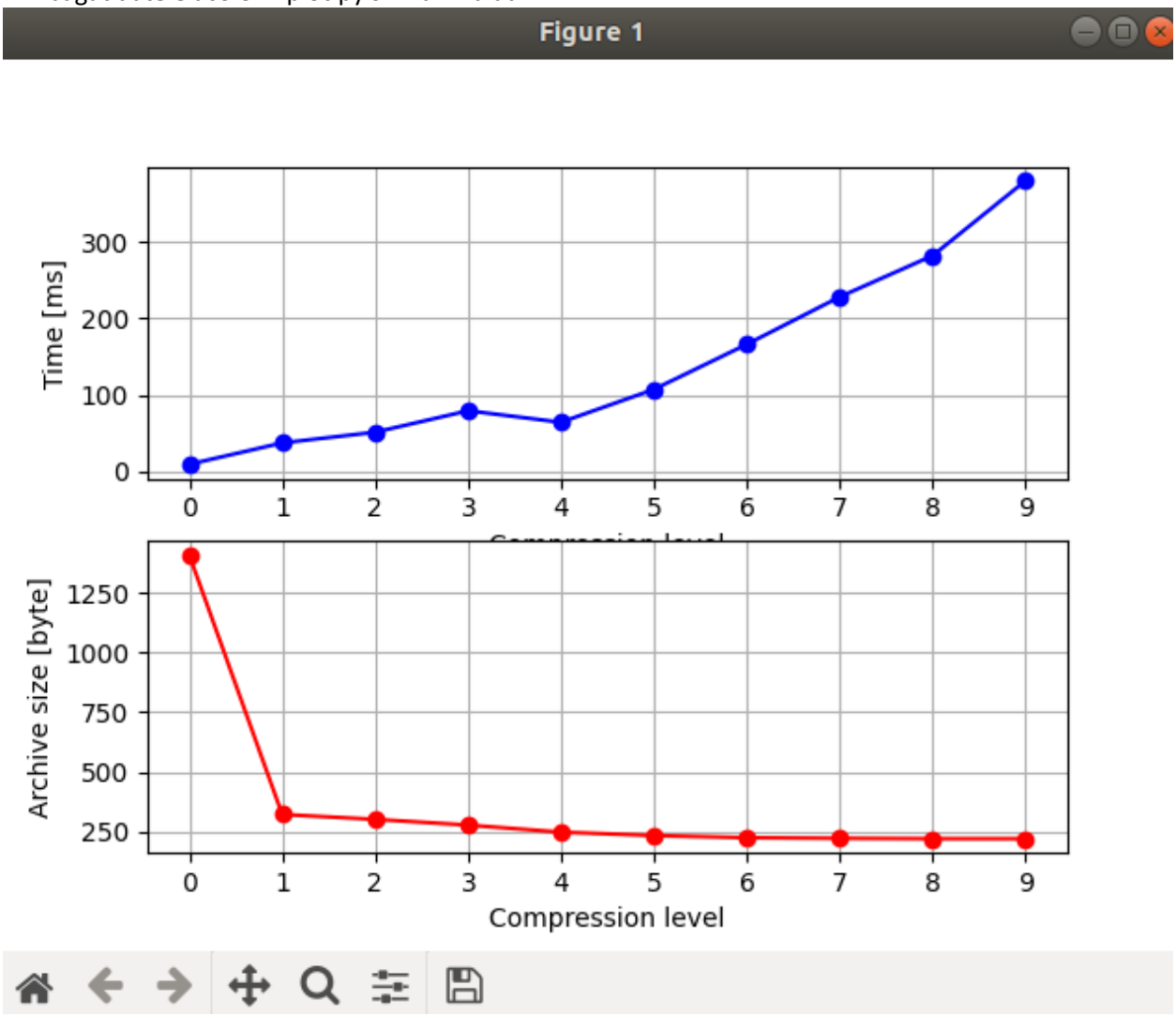
real    0m0,488s
user    0m0,191s
sys     0m0,000s

real    0m0,382s
user    0m0,208s
sys     0m0,000s


student@uso:~/Desktop/lab1$ ls -l
total 15632
-rw-rw-r-- 1 student student 4264316 oct 19 20:17 MARBLES.BMP
```

-rw-rw-r-- 1 student student 1440054 oct 19 20:18 RAY.BMP
-rw-rw-r-- 1 student student 6488666 oct 19 20:17 big.txt
-rwxr-xr-x 1 student student 353 oct 20 10:33 fact_rcrs.py
drwxr-xr-x 2 student student 4096 iul 26 06:20 hw_counter
-rwxr-xr-x 1 student student 869 oct 20 11:17 plot.py
-rw-r--r-- 1 student student 1440218 oct 20 11:20 ray-0.zip
-rw-r--r-- 1 student student 331253 oct 20 11:20 ray-1.zip
-rw-r--r-- 1 student student 310248 oct 20 11:20 ray-2.zip
-rw-r--r-- 1 student student 285757 oct 20 11:20 ray-3.zip
-rw-r--r-- 1 student student 255569 oct 20 11:20 ray-4.zip
-rw-r--r-- 1 student student 240879 oct 20 11:20 ray-5.zip
-rw-r--r-- 1 student student 231742 oct 20 11:20 ray-6.zip
-rw-r--r-- 1 student student 229192 oct 20 11:20 ray-7.zip
-rw-r--r-- 1 student student 227524 oct 20 11:20 ray-8.zip
-rw-r--r-- 1 student student 227390 oct 20 11:20 ray-9.zip

Am bagat datele acolo in plot.py si l –am rulat



TASK 4

Nu am linux nativ, nu mi-a mers pe masina virtuala sau pe subsistem sa fac taskul

Acesta este codul scris, insa nu il pot rula

```
/* TODO: count L2 cache misses for the next block using RDPMC */
            rdpmc(ecx, eax, edx);
            counter = ((uint64_t) eax) | (((uint64_t) edx) << 32);

   /* perform slow multiplication */
   for (uint32_t i=0; i<N; ++i)           /* line   */
      for (uint32_t j=0; j<N; ++j)        /* column */
         for (uint32_t k=0; k<N; ++k)
            r[i*N + j] += m1[i*N + k] * m2[k*N + j];
            rdpmc(ecx, eax, edx);
            counter = (((uint64_t) eax) | (((uint64_t) edx) << 32)) - counter;

   /* TODO: count L2 cache misses for the next block using RDPMC */
rdpmc(ecx, eax, edx);
            counter = ((uint64_t) eax) | (((uint64_t) edx) << 32);

   /* perform fast(er) multiplication */
   for (uint32_t k=0; k<N; ++k)
      for (uint32_t i=0; i<N; ++i)        /* line   */
         for (uint32_t j=0; j<N; ++j)     /* column */
            r[i*N + j] += m1[i*N + k] * m2[k*N + j];
rdpmc(ecx, eax, edx);
counter = (((uint64_t) eax) | (((uint64_t) edx) << 32)) - counter;
```

TASK 5

